Lab 03 Tasks

A student needs a simple calculator to perform basic mathematical operations.

Requirements:

1. Ask the user to enter two numbers.
2. Ask the user to select an operation:
    - Addition
    - Subtraction
    - Multiplication
    - Division
3. Display the calculated result.
4. Handle division by zero

A teacher wants to save students' marks so that the record can be viewed later.

Requirements:

1. Ask the user for the student's name.
2. Enter marks of 5 subjects.
3. Calculate total and percentage.
4. Save the complete record in a .txt file.
5. Read the file and display the saved information.

Example file:

Student Name: Ali
Programming: 85
Networking: 78
Database: 90
Mathematics: 82
English: 75
Total: 410
Percentage: 82%

A student wants to send a short message to a department server.

Requirements:

Server:

1. Create a TCP socket.
2. Bind and listen.
3. Accept a client.
4. Receive a message.
5. Display the message.

Client:

1. Connect to the server.
2. Ask the user for a message.
3. Send the message.
4. Close the connection.

A university maintains a simple server that receives student information.

Requirements:

1. Client asks the user for:
   - Student ID
   - Student name
   - Department
2. Send the information to the server.
3. Server receives and displays the information.
4. Server sends an acknowledgment to the client.

A network administrator has a list of ports discovered during troubleshooting and wants to identify their common services.

Requirements:

1. Store at least 10 port numbers in a list.
2. For each port:
   - Determine the TCP service name.
   - Determine the UDP service name where available.
3. Display the results in a table.
4. Handle unknown services.

   Example:

```
Port    TCP Service    UDP Service
---------------------------------------------
21      ftp            ftp
22      ssh            ssh
53      domain         domain
80      http           http
```

A university wants to generate a complete semester GPA report for a student.

Requirements:

For each course, input:

- Course name
- Credit hours
- Letter grade

Calculate:

- Grade point
- Quality points
- Total credit hours

- Total quality points
- Semester GPA

Display the report in the following format:

```
Course        CH      Grade      GP      Quality Points
-------------------------------------------------------
Programming   3       A          4.0     12.0
Networks      3       B+         3.33    9.99
Database      3       A-         3.67    11.01
-------------------------------------------------------
Total         9                  33.00

Semester GPA = 3.67
```

A university wants to store student information in a structured format instead of a plain text file.

Requirements:

1. Enter:
   - Student ID
   - Name
   - Department
   - Semester
   - GPA
2. Store the information in a .json file.
3. Read the JSON file.
4. Display the student information.

Example JSON:

```
{
  "student_id": "2026-CS-101",
  "name": "Ali",
  "department": "Computer Science",
  "semester": 3,
  "gpa": 3.52
}
```